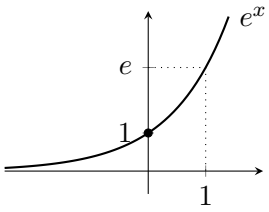
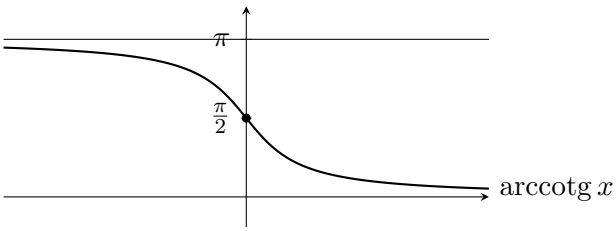
Z grafu funkce určete limity:

a) $\lim\limits_{x\to\infty} e^x = \infty$, $\lim\limits_{x\to-\infty} e^x = 0$, $\lim\limits_{x\to 0} e^x = 1$,

$\lim\limits_{x\to 1} e^x = e$



b) $\lim\limits_{x\to\infty} \operatorname{arccotg} x = 0$, $\lim\limits_{x\to-\infty} \operatorname{arccotg} x = \pi$, $\lim\limits_{x\to 0} \operatorname{arccotg} x = \frac{\pi}{2}$, $\lim\limits_{x\to 1} \operatorname{arccotg} x = \operatorname{arccotg} 1 = \boxed{\frac{\pi}{4}}$



Vypočtěte:

a)

$$\lim_{x\to\infty} e^{\frac{2x}{x^2-1}} = e^0 = \boxed{1} \qquad (složená\ funkce)$$

$$\left(\lim_{x\to\infty} \frac{2x}{x^2-1} = \lim_{x\to\infty} \frac{x\cdot 2}{x\left(x-\frac{1}{x}\right)} = \frac{2}{\infty} = 0\right)$$

b)

$$\lim_{x\to\infty} \frac{3x^4-2}{2x^4+x+3} = \lim_{x\to\infty} \frac{\cancel{x^4}\left(3-\overbrace{\frac{2}{x^4}}^{\to 0}\right)}{\cancel{x^4}\left(2+\underbrace{\frac{1}{x^3}}_{\to 0}+\underbrace{\frac{3}{x^4}}_{\to 0}\right)} = \boxed{\frac{3}{2}}$$

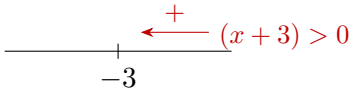
c)

$$\lim_{x\to -3^+} \frac{2-x}{2x^2+5x-3} = \left[\frac{5}{0}\right] = \lim_{x\to -3^+} \frac{2-x}{2(x+3)\left(x-\frac{1}{2}\right)} =$$

$$D = 25-4\cdot 2\cdot(-3), \quad D = 49 \;\Rightarrow\; x_{1,2} = \frac{-5\pm 7}{4} = \begin{cases} -3 \\ \frac{2}{4} = \frac{1}{2} \end{cases}$$

$$= \lim_{x\to -3^+} \frac{1}{x+3} \cdot \lim_{x\to -3^+} \frac{2-x}{2\left(x-\frac{1}{2}\right)} = \frac{1}{0^+} \cdot \frac{5}{2\left(-3-\frac{1}{2}\right)} =$$

$$= \infty \cdot \frac{5}{2\cdot\left(-\frac{7}{2}\right)} = \infty\cdot\left(-\frac{5}{7}\right) = \boxed{-\infty}$$



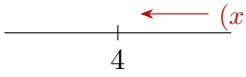
d)

$$\lim_{x\to 4^+} \ln \frac{3x+1}{x^2-4x} = \ln \text{„}\infty\text{“} = \infty \qquad \left(\lim_{x\to\infty} \ln x = \infty\right)$$

$$\lim_{x\to 4^+} \frac{3x+1}{x^2-4x} = \lim_{x\to 4^+} \frac{3x+1}{x(x-4)} = \lim_{x\to 4^+} \frac{1}{x-4} \cdot \lim \frac{3x+1}{x} =$$

$$= \frac{1}{0^+} \cdot \frac{3\cdot 4+1}{4} = \infty \cdot \frac{13}{4} = \boxed{\infty}$$

$(x-4) > 0$ ⟵



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e)

$$\lim_{x\to-\infty} \left(x^3 - x + 2\right) = \lim_{x\to-\infty} x\Big(x^2 - 1 + \underbrace{\tfrac{2}{x}}_{\to 0}\Big) =$$

$$= -\infty \cdot (-\infty)^2 = -\infty \cdot (+\infty) = \boxed{-\infty}$$

!! $\boxed{\lim_{x\to-\infty} x^3 - x + 2} = (-\infty)^3 - x + 2 = -\infty - x + 2$ (lim jen pro první člen)

Špatný zápis, pokud chceme limitu z celého výrazu $(x^3 - x + 2)$.